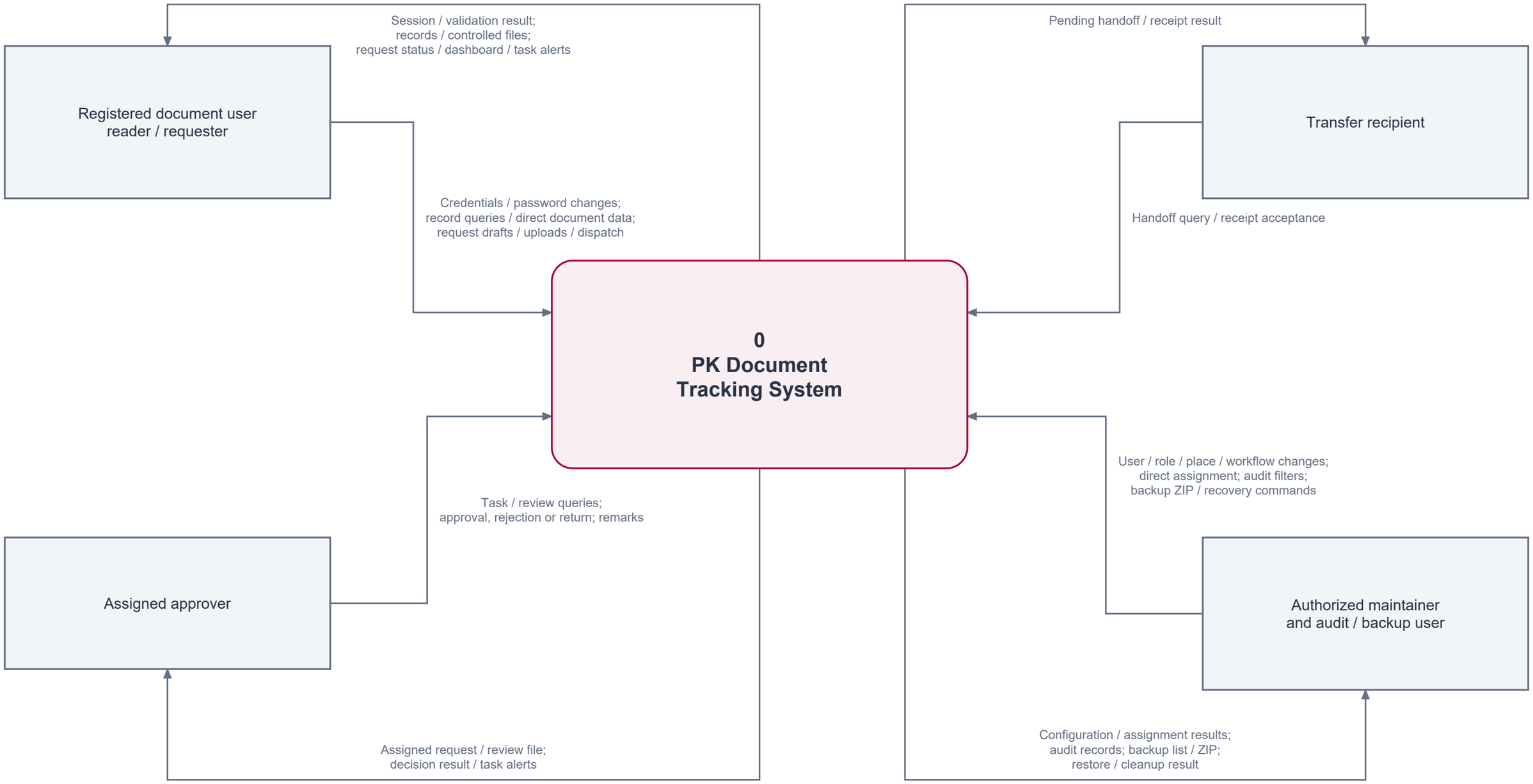
Proposed system context

Proposed participants, system boundary and data exchanges

External participant | Process / activity | Arrows: data flow

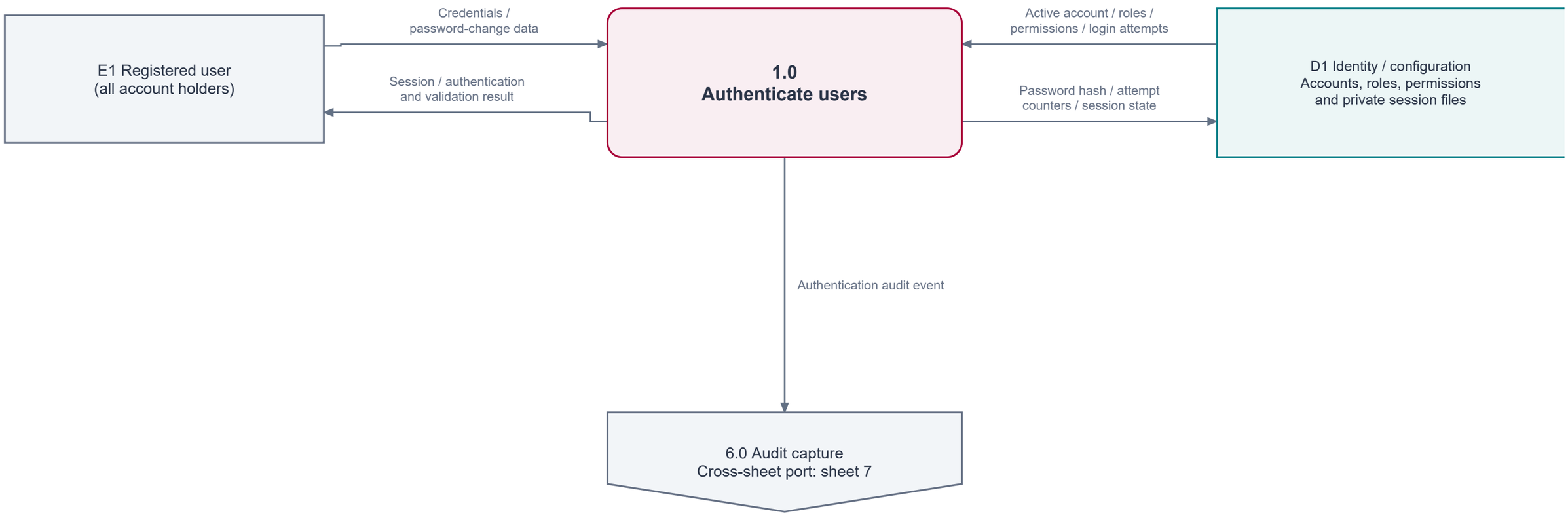


- Actors represent proposed responsibilities. An authorized account may perform several responsibilities; published workflows will select the approvers.
- Dispatch and receipt confirmations will cross the software boundary as data. Physical paper will move between people outside that boundary.
- All participants will sign in as registered users. Dashboard and task alerts will be delivered within the application.
- Sheets 2-7 define six Level 0 processes using consistent identifiers and shared logical stores. Sheet 8 specifies the proposed stores.

Level 0 / 1.0 - authenticate users

Credentials will establish a checked identity and session for each registered participant

External participant | Process / activity | Data store | Cross-sheet reference | Arrows: data flow

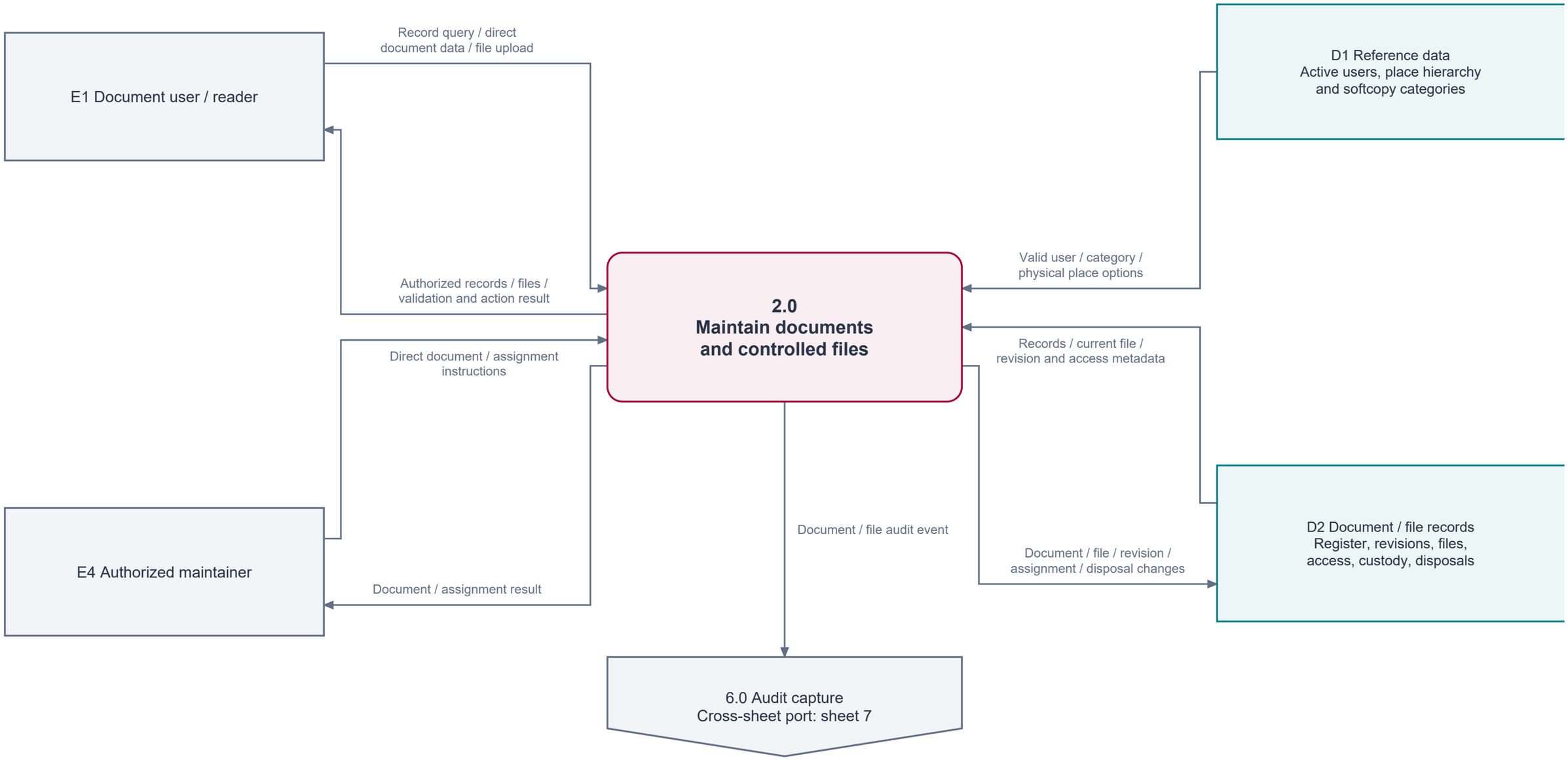


- Active accounts will receive the combined permissions of their active roles. The Administrator role will provide system-wide permission.
- Authentication will verify passwords, check login attempts and establish sessions. Required password setup shall finish before ordinary protected actions.
- D1-D4 identify the logical store groups specified on sheet 8. The audit port links to proposed process 6.0 on sheet 7.

Level 0 / 2.0 - maintain documents and files

Authorized changes, scoped browsing and controlled file delivery will use a shared document register

External participant | Process / activity | Data store | Cross-sheet reference | Arrows: data flow

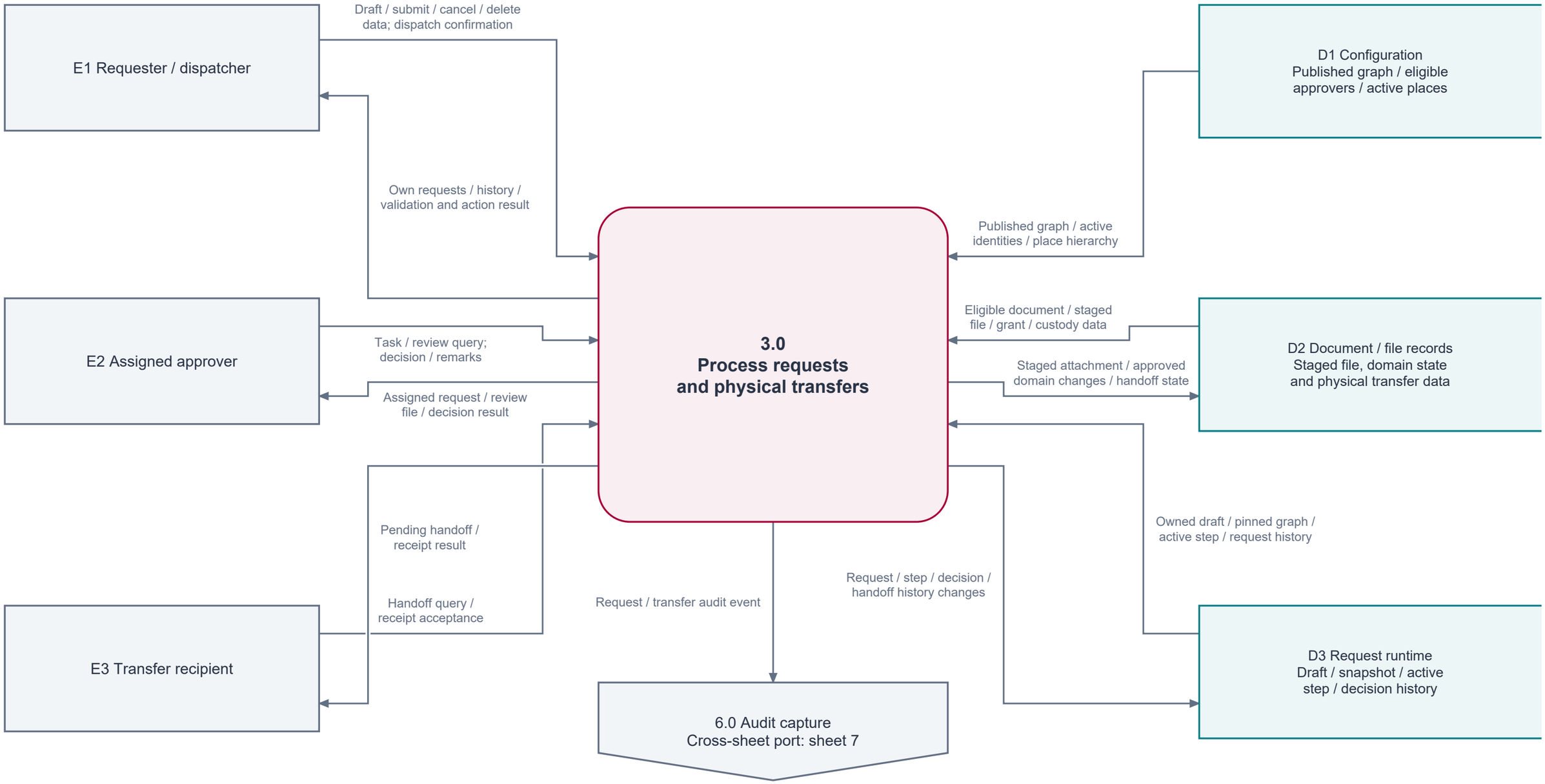


- Document browsing, direct maintenance and controlled file delivery will check the user permission and document scope.
- Softcopy assignment will add an assignment record. Hardcopy assignment will change the current holder.
- File review, download and printing will use protected routes. Optional Office/WPS review will open the local editor through a signed content link.
- D1-D4 identify the logical store groups on sheet 8. The audit port links to proposed process 6.0 on sheet 7.

Level 0 / 3.0 - process requests and transfers

Drafts and decisions will produce captured approval routes, document changes and custody records

External participant | Process / activity | Data store | Cross-sheet reference | Arrows: data flow

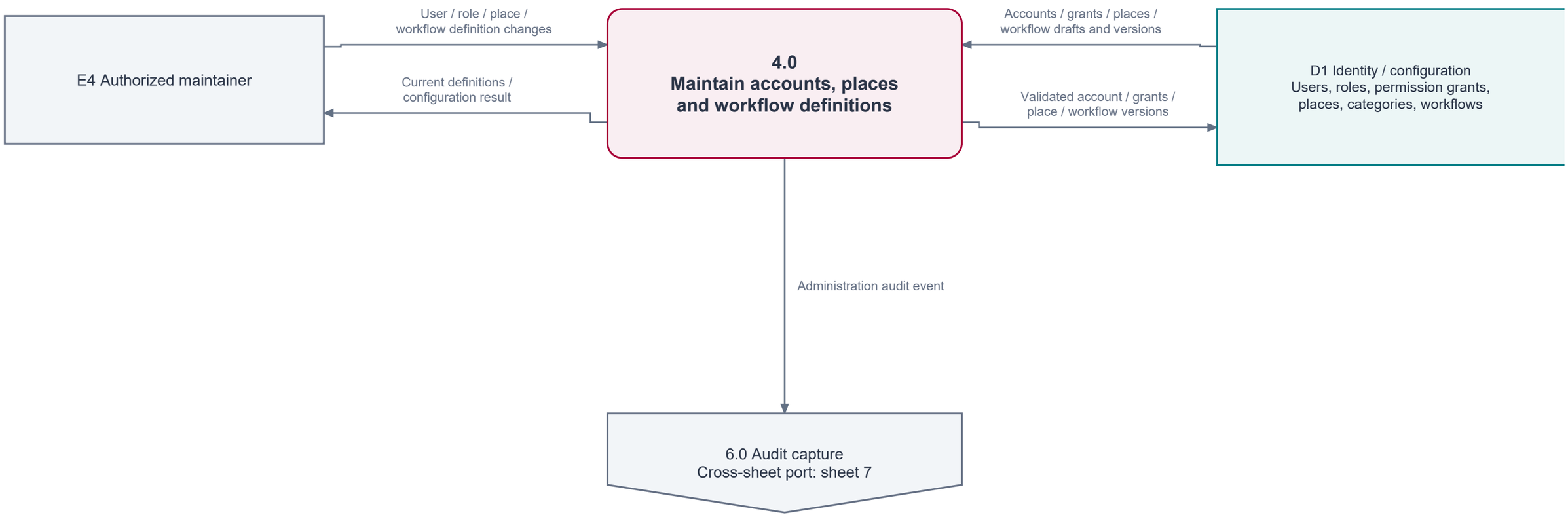


- Nine request types will cover softcopy creation/revision/cancellation, hardcopy creation/update, transfer, disposal, access and assignment.
- Submission will capture a published route. Approve, Reject and Return will update the active step and history; final approval will apply the selected operation.
- Only the transfer requester will dispatch. Recipient acceptance will change custody/location, remove the previous personal grant and complete the request.
- Only the owner will be allowed to delete a cancelled, never-submitted draft. D1-D4 are specified on sheet 8; audit capture links to process 6.0 on sheet 7.

Level 0 / 4.0 - maintain system definitions

Validated definitions will govern account access, document locations and approval routes

External participant | Process / activity | Data store | Cross-sheet reference | Arrows: data flow

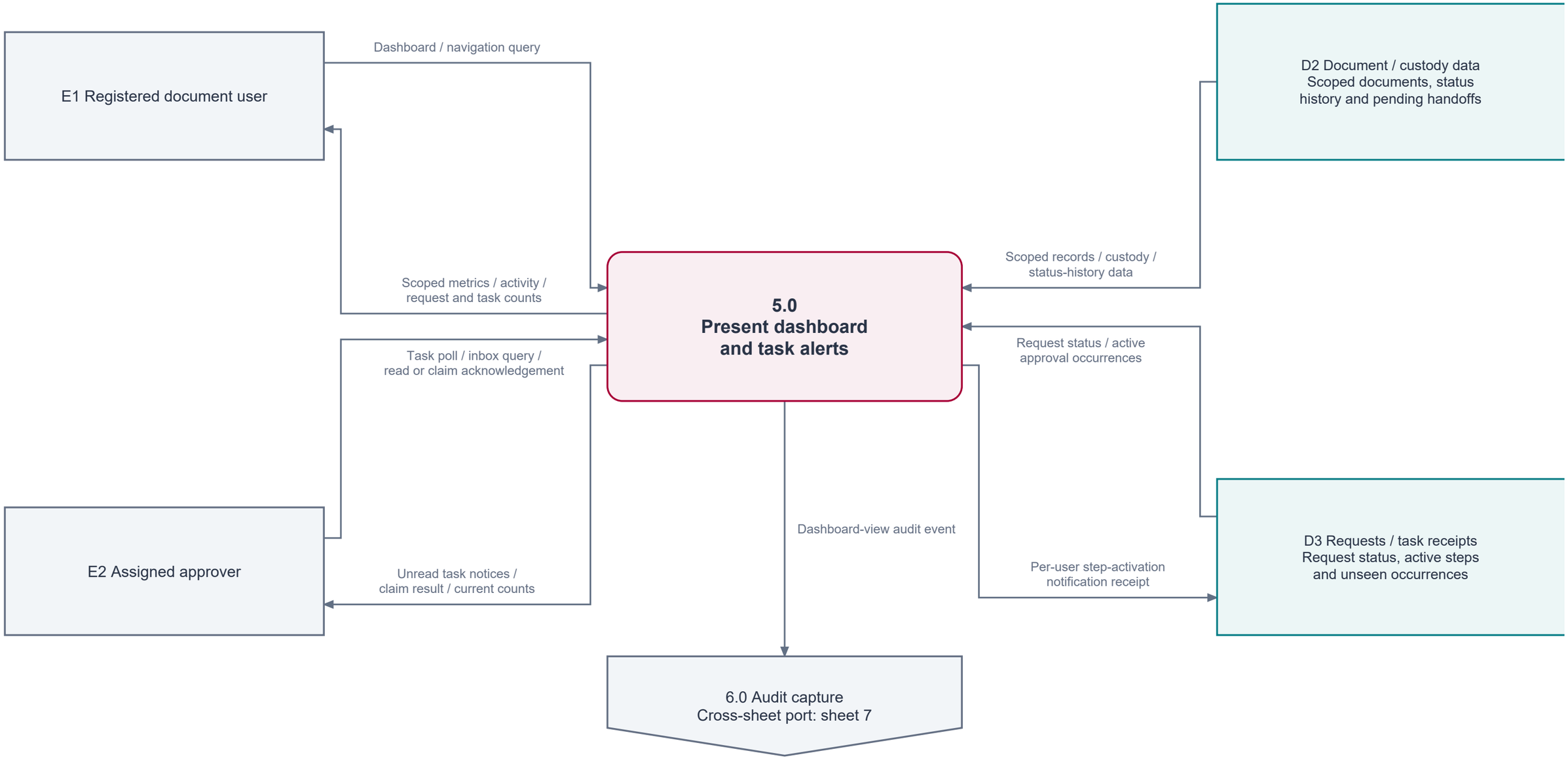


- An account may hold several active roles. User and role maintenance will enforce delegation rules and refresh affected sessions.
- Hardcopy places will follow Area > Specific > optional Asset > Location. Softcopy categories may contain subcategories.
- Published workflows will be versioned. Editing a clone will create a separate draft; submitted requests will retain their captured approval route.
- D1-D4 identify the logical store groups on sheet 8. The audit port links to proposed process 6.0 on sheet 7.

Level 0 / 5.0 - present dashboard and task alerts

Scoped records and active approvals will supply dashboard metrics and application task notices

External participant | Process / activity | Data store | Cross-sheet reference | Arrows: data flow

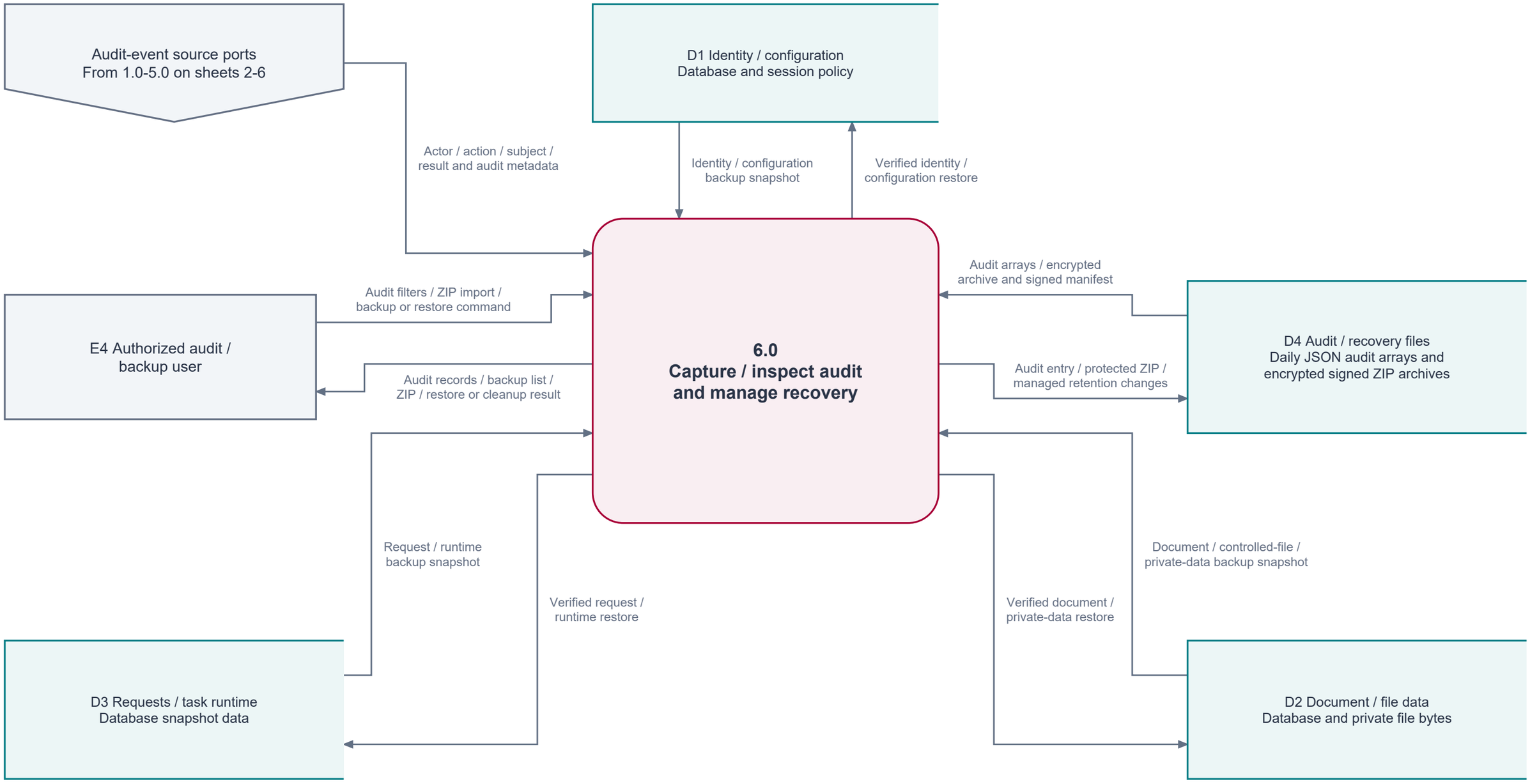


- Dashboard counts and recent activity will follow the signed-in user document scope. Approval notices will follow the exact active step assignment.
- Task notices will use request state, workflow steps and per-user occurrence receipts. Claiming a notice will prevent repeated opening across tabs or devices.
- D1-D4 identify the logical store groups on sheet 8. The audit port links to proposed process 6.0 on sheet 7.

Level 0 / 6.0 - capture audit and manage recovery

Operational events will form private audit records; protected snapshots will support recovery

External participant | Process / activity | Data store | Cross-sheet reference | Arrows: data flow



- Audit events will be stored in private daily JSON files. Audit listing, backup creation and restore will require their respective permissions.
- Backups will contain a SQL dump plus application and private files, excluding sessions, cache and backup recursion. AES-256 will encrypt entries; a signed manifest will verify contents.
- Restore will first create a recovery snapshot, then reapply database and private working data. Maintenance will block ordinary requests during restore. Application-code recovery will be offline.
- Retention will manage stored archives and daily audit/application logs. D1-D4 are specified on sheet 8.

Proposed data store catalog

Proposed physical structures for the D1-D4 logical stores referenced on sheets 2-7

Data store

D1 Identity and configuration
SQL: users, roles, user_roles, role_permissions, permissions,
login_attempts; areas, specifics, assets, locations, categories;
workflows, workflow_versions
Private session files: storage/sessions

D2 Document, file and custody records
SQL: hardcopy_documents, softcopy_documents, softcopy_revisions,
status_history, disposals, files, assignments, access_grants, transfers
Controlled file bytes: storage/documents/<random storage_name>
Assignment storage: assignments for softcopies; holder_id for hardcopies.

D3 Request and approval runtime
SQL: requests, workflow_steps, workflow_history, task_notification_receipts
Stored data: proposal payload, pinned workflow/version, active assignments,
decision history and per-user task occurrence receipts

D4 Audit and recovery files
Daily audit JSON arrays: storage/audit/YYYY-MM-DD.json
Encrypted AES-256 ZIP archives: storage/backups (or configured safe backup path)
Backup contents: signed manifest, SQL snapshot and application and private files.
Retention scope: managed backups, daily audit and application logs.

- D1-D4 specify related persistent data for the processes on sheets 2-7. SQL table names and private storage locations define the proposed physical structures.
- Roles, permissions and approval assignments will be configurable. The task inbox will use request state, active steps and per-user occurrence receipts.
- Audit will use private daily files. Backups will combine a database snapshot with application and private files; web restore will recover database and private working data.
- Optional Office/WPS review will open the local editor through a signed content link.